

Chapter 30

Use of Microbial Microarrays to Define Antibody Specificity

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Abstract

The formation of alloantibodies against red blood cell (RBC) antigens is the most frequent immune barrier to blood transfusion. Among these antibodies, those that occur naturally within the first few months of life against carbohydrate-based ABO blood group antigens are the most prevalent. Individuals who lack the A or B antigen produce antibodies directed against the corresponding absent antigen. Unlike antibodies that arise following antigen exposure during pregnancy or transfusion, these spontaneously occurring antibodies suggest a role for environmental influences. Early investigations proposed that microbial pathogens bearing carbohydrate motifs resembling blood group antigens could provide such a stimulus. However, characterizing antibody specificities against diverse microbes remains difficult, as each organism often requires unique growth conditions. To address this challenge, glycan microarray technology has been developed, which has enabled high-throughput assessment of antibody binding to a wide variety of microbial carbohydrate determinants, including those mimicking blood group antigens. In this work, we highlight the use of microbial glycan microarrays for defining specificity of antibodies and lectins against microbial mimics of blood group and related antigens.

Key words Immunity, Galectin, Antibody, MGM microarray, Molecular mimicry

1 Introduction

Transfusion of blood products remains one of the most common interventions in hospitalized patients worldwide. The need for transfusion arises in a wide range of clinical contexts, including surgical blood loss, traumatic injury, hematologic disorders, and as critical form of supportive care in oncology [1, 2]. For transfusions to be performed safely, compatibility between donor and recipient blood must be established [3–5]. This requirement stems from the fact that individuals naturally generate antibodies against polymorphic carbohydrate structures expressed on red blood cells (RBCs), most notably the ABO blood group antigens [6–8]. Incompatible transfusions can lead to the rapid opsonization and destruction of donor RBCs, provoking severe inflammatory

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responses that may culminate in life-threatening hemolytic transfusion reactions [9–13].

In addition to these naturally occurring anti-ABO antibodies, alloimmunization can also occur when individuals are exposed to additional non-self RBC alloantigens that often reflect protein-based polymorphisms expressed on the RBC surface [14–19]. Such antibodies are typically induced following transfusion or during pregnancy as a result of fetomaternal hemorrhage [20–25]. These immune responses are often directed against minor blood group antigens, where the prevalence of these antibodies is much lower than antibodies that naturally arise against ABO antigens [6, 24, 26, 27]. However, despite antigen targets being designated as minor simply to reflect their comparative frequency to ABO blood group antigens, antibodies against these RBC antigens can have major clinical impact if an incompatible transfusion occurs, leading to equally severe hemolytic transfusion reactions and associated complications [9, 28–32].

Although ABO blood group antigens were first described more than a century ago, the immunological mechanisms driving the formation and overall consequences of both naturally occurring anti-ABO antibodies and RBC-induced alloantibodies remain incompletely defined [17, 33–40]. Emerging studies suggest that alloantibody induction after RBC exposure involves the activation of innate-like B cells, as well as innate immune signaling pathways possibly being driven by reticulocytes present in the donated unit that promote antibody production even in the absence of conventional adjuvants [41–50]. Intriguingly, analogous immune mechanisms have been proposed for alloantibody responses against platelet antigens and for inhibitors that develop against therapeutic clotting factors [51–53]. Although antibody-independent mechanisms for platelet clearance have been reported [54], suggesting additional mechanisms may be involved, prior studies examining alloantibody induction raise the possibility that conserved pathways underlie immune responses to intravenously administered biologics [44, 47, 51, 52, 55]. Additional studies have explored mechanisms underlying the consequences of antibody engagement of RBCs in vivo [28–30, 37, 38, 56–59].

In contrast, the spontaneous generation of anti-ABO antibodies may be environmentally driven, although alternative hypotheses certainly exist [33, 60, 61]. A prevailing hypothesis is that antigenic mimicry by microbes plays a central role, as many bacterial species decorate their surfaces with glycan structures that resemble mammalian blood group determinants [62]. Repeated exposure to such cross-reactive microbial glycans during early life may underlie the development of naturally occurring antibodies of direct relevance to both transfusion and transplantation medicine [63]. However, dissecting these interactions requires tools capable of assessing glycan recognition with structural resolution.

To address this, microbial glycan microarrays (MGMs) have been developed that are populated with hundreds of structurally diverse glycans derived from a wide array of microbial species [64–70], similar to strategies employed to define the binding specificity of antibodies and lectins toward mammalian glycan structures, including blood group antigens [71–82]. This high-throughput platform enables simultaneous assessment of antibody and lectin binding against diverse glycan repertoires, thereby providing unique insights into host–microbe interactions that may influence the development of naturally occurring anti-blood group antibodies [65]. We will describe, herein, the use of MGMs to characterize the specificity of antibodies and lectins that target microbial mimics of blood group antigens.

2 Materials

2.1 Culturing and Drying of Bacteria

Special Equipment:

1. Lyophilizer (Labconco).
2. Spectrophotometer (capable of reading OD_{600}).

2.1.1 Medias

1. Luria Bertani (LB) culture media (Fisher).
2. Brain Heart Infusion Media (Bacto, Becton Dickinson, Franklin Lakes, NJ).
3. Hemin, 10 mg/mL (Sigma Aldrich, St. Louis, MO).
4. NADH, 10 mg/mL (Sigma Aldrich).
5. Todd Hewitt Broth (THB) (Bacto, Becton Dickinson, Franklin Lakes, NJ).

2.1.2 Bacteria Cultured in LB Culture Media

1. *E. coli* O86:B7.
2. *S. marcescens* O20.
3. *S. marcescens* control strain (*ATCC#14756*).

2.1.3 Bacteria Cultured in THB

1. *S. agalactiae* A909 (*ATCC#BAA-1138*).

2.2 Preparation of Bacterial Polysaccharides (BPS)

1. Dried bacterial cells (*see* Subheading 2.1 Culturing of bacteria).
2. 90% phenol in water.
3. DNAse, 50 μg/mL.
4. RNAse, 50 μg/mL.
5. Ultracentrifuge machine.
6. 2% acetic acid.
7. Sephadex G-50 column.
8. 0.05 M pyridinium acetate buffer, pH 4.5.

9. Printing buffer: 300 mM phosphate buffer, 0.005% Tween-20, pH 8.4.
10. 96-well microplates.
11. BioMek 1000 printing robot (Beckman/Coulter).

2.3 Printing of the Microbial Glycan Microarray (MGM)

1. Diluted glycan antigens.
2. 384-well microplates.
3. Pipetting robot (BioMek 2000, Beckman Coulter).
4. MicroGrid II (Digilab) microarray printing robot equipped with solid steel Stealth microarray pins (SMP4B, TeleChem).
5. SlideH (Schott/Nexterion) or CodeLink (GE Healthcare) NHS-ester activated microscope slides.
6. Desiccator.
7. Blocking buffer: 50 mM ethanolamine in 50 mM borate buffer.
8. Double distilled water.

2.4 Analysis of Binding to the MGM

1. Sera of humans (*see* **Notes 1** and **4**).
2. Sera of mice, rabbits (*see* **Notes 2** and **4**).
3. Intravenous immunoglobulin (IVIG) (*see* **Notes 3** and **4**).
4. Humidified chamber (*see* **Note 5**).
5. TSM Binding Buffer: Tris buffer/Salts/Metal ions (TSM)—20 mM Tris–HCl, 150 mM sodium chloride, 2 mM calcium chloride, 2 mM magnesium chloride, pH 7.4.
6. Alexa 488-streptavidin (Invitrogen, cat. no. S32354).
7. Goat anti-human IgG, Alexa Fluor 488 (Invitrogen, cat. no. A11013).
8. Goat anti-human IgM, Alexa Fluor 488 (Invitrogen, cat. no. A21215).
9. Goat anti-mouse IgG, Alexa Fluor 633 (Invitrogen, cat. no. A21050).
10. Goat anti-mouse IgM, Alexa Fluor 488 (Invitrogen, cat. no. A21042).
11. Goat anti-rabbit IgG, Alexa Fluor 488 (Invitrogen, cat. no. A11034).
12. Goat anti-human IgG, Alexa Fluor 488 (Invitrogen, cat. no. A21091).
13. Goat anti-mouse IgG, Alexa Fluor 488 (Invitrogen, cat. no. A11001).
14. Microarray scanner (Scan Array Express, PerkinElmer Life Sciences).
15. Imagene software (BioDiscovery) (Fig. 1).

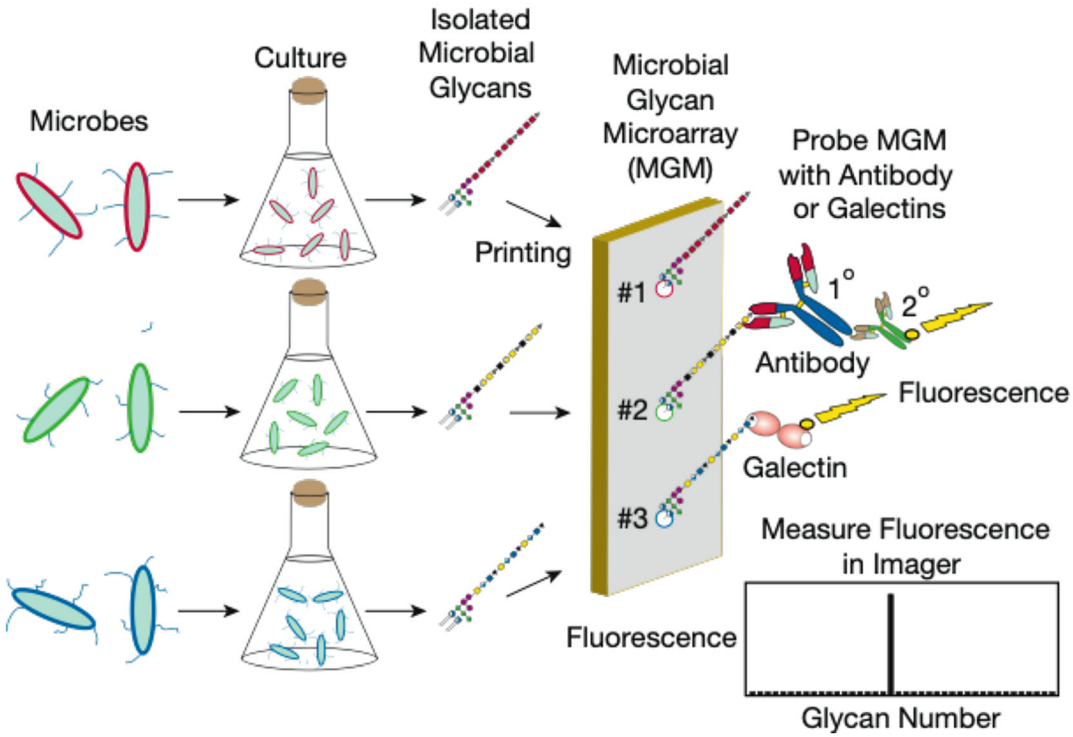


Fig. 1 Illustrates the process of generating microbial glycan microarrays (MGMs). Initially, bacterial cultures undergo amplification, after which individual surface glycans are isolated. These glycans are subsequently attached to slides to generate MGMs, whice can then be probed with various immune factors, such as antibodies in sera. (This research was originally published in Nature Chemical Biology. Stowell et al. [70])

3 Methods

3.1 Culturing and Drying of Bacteria

1. Grow and maintain *E. coli* O86:B7, *S. marcescens* O20, and *S. marcescens* control strain ATCC #14756 in aerobic conditions at 37 °C using LB culture media.
2. Grow and maintain *S. agalactiae* A909 in standard aerobic conditions at 37 °C using Todd Hewitt Broth (THB).
3. Monitor culture growth by OD_{600} and harvest the bacterial cultures between OD_{600} 0.6–0.8 by centrifugation to obtain cell pellets.
4. Wash the cell pellets with PBS to remove residual growth medium components.
5. Lyophilize (freeze-dry) the washed cell pellets to obtain dried bacterial cells.

3.2 Preparation of Bacterial Polysaccharides (BPS)

1. Grind 1 g lyophilized bacteria pellet into a fine powder.
2. Resuspend 1 g of pellet in 5 mL of 50 mM sodium hydrogen phosphate (pH 7) with 5 mM EDTA. This buffer was prepared by dissolving 3.5 g sodium phosphate and 0.93 g EDTA in 500 mL dH_2O. This solution can be preserved at room temperature.
3. Vortex until fully dissolved, then leave on a shaker to help dissolve any clumps that might be left.
4. Add lysozyme and leave on a shaker at room temperature overnight.
5. Place the resuspended pellet with lysozyme in a 37 °C water bath for 20 min.
6. Add 15 mL of 50 mM Na_2HPO_4 with 20 mM $MgCl_2$. This buffer was prepared by dissolving 3.5 g Na_2HPO_4 and 2 g $MgCl_2$ in 500 mL dH_2O. This solution can be preserved at room temperature.
7. Add DNAse and RNAse, both at 50 μg/mL.
8. Incubate in a 37 °C water bath for 1 h.
9. Move to a 60 °C water bath for 1 h.
10. Preheat 90% phenol (18 g phenol +2 g H_2O) to 70 °C (*see* **Note 6**).
11. Mix phenol with bacteria in equal volumes after both are equilibrated to 70 °C.
12. Leave for about 10 min in the water bath.
13. Remove and cool quickly in an ice water bath.
14. Centrifuge at about 3000× g for 40 min.
15. Take the upper (aqueous phase) layer and dialyze against dH_2O until the phenol odor disappears.
16. Dialyze LPS with 4 L changes of dH_2O for 4 days.
17. Recover about 30 mL of dialyzed material (*see* **Note 7**).
18. If bacterial polysaccharides (BPs) are released from lipopolysaccharide (LPS), subject the isolated LPS to mild acid hydrolysis using 2% acetic acid (v/v) to remove the lipid A portion, leaving the inner core, outer core, and O-antigen (BPs).
19. Fractionate the isolated supernatants using a Sephadex G-50 column in 0.05 M pyridinium acetate buffer at pH 4.5.
20. Aliquot the samples and lyophilize for storage.
21. Prior to printing microbial glycans, dissolve lyophilized samples in printing buffer (300 mM phosphate buffer plus 0.005% Tween-20, pH 8.4) to a stock concentration of 1 mg/mL.
22. Dilute the stock solution into 96-well microplates prior to printing by performing twofold serial dilutions using a BioMek 1000 printing robot (Fig. 2).

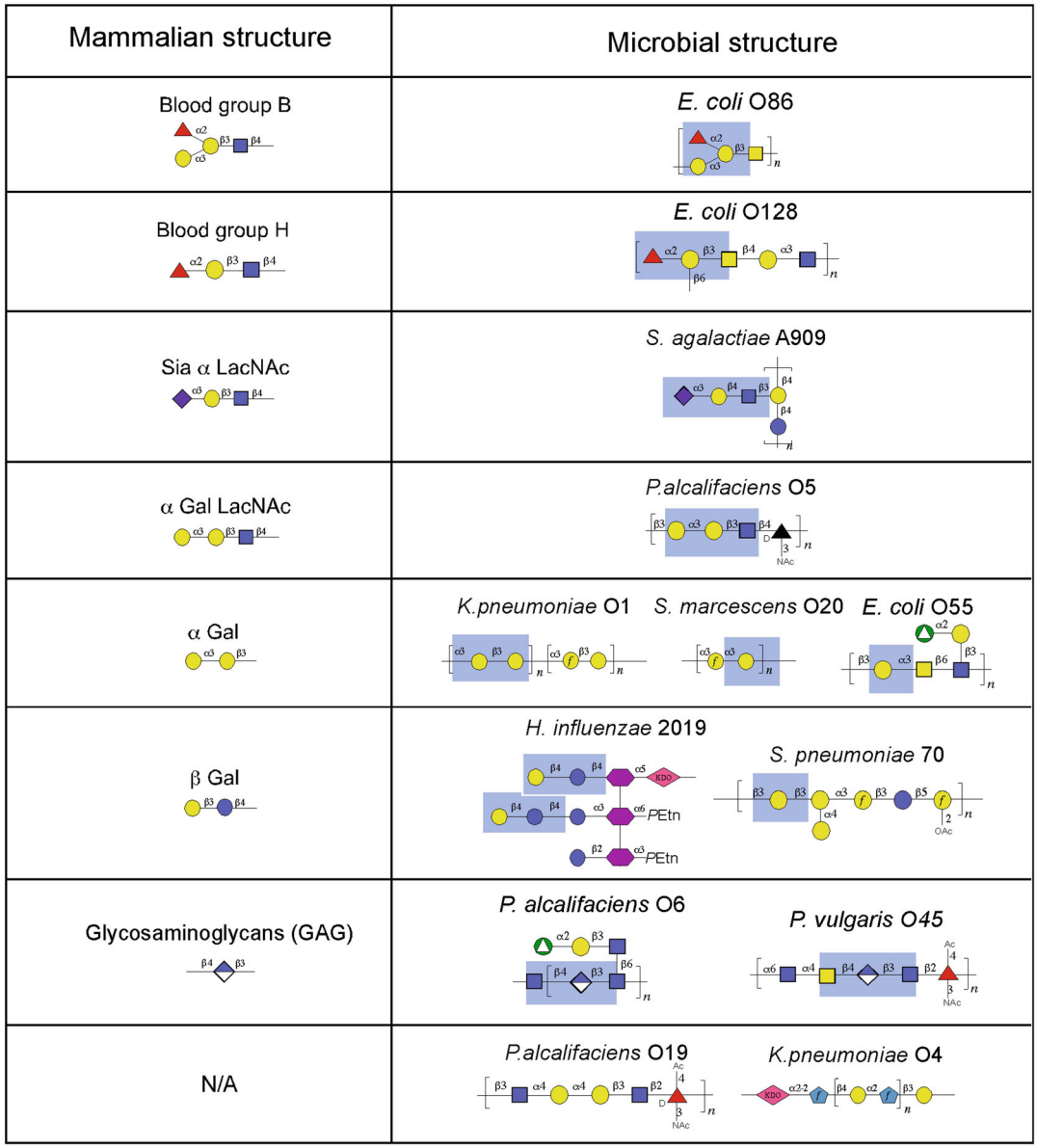


Fig. 2 Schematic representation of mammalian and microbial carbohydrate determinants. Structures of commonly occurring mammalian glycans are shown as indicated with corresponding structures synthesized by the indicated microbial species shown on the right. The carbohydrate symbols used in this figure are defined in the expanded carbohydrate symbol nomenclature shown in Fig. 3. (This research was originally published in Nature Chemical Biology. Stowell et al. [70])



Fig. 3 Symbol nomenclature. (This research was originally published in Nature Chemical Biology. Stowell et al. [70])

3.3 Printing of the Microbial Glycan Microarray (MGM)

1. Deposit 10 μL of diluted glycan antigens into 384-well microplates using a pipetting robot. Then, spot them onto SlideH or CodeLink NHS-ester activated microscope slides with a microarray printing robot equipped with solid steel Stealth microarray pins.
2. Each pin delivers ~1 nL of dissolved sample per spot deposition and six replicate spots of each sample should be imprinted on each array.
3. Print arrays at ~55% relative humidity (RH) and allow 30 min of immobilization at 100% RH before being desiccated at RT. C.
4. After desiccation, immerse the printed arrays in the blocking buffer (50 mM ethanolamine in 50 mM borate buffer) for 1 h on a rotary shaker.
5. Following blocking, rinse slides in ddH_2O and dry by centrifugation prior to incubation/analysis.

3.4 Analysis of Binding to the MGM

1. Printed MGMs were assessed for functionality using sera from humans, mice, and rabbits, as well as IVIG. The sera analyzed on the array represent pooled samples from five individual donors.
2. MGM data were obtained after incubation with pooled normal human sera at a 1:1000 dilution, followed by labeling with anti-IgG and anti-IgM antibodies. Alternatively, the samples were probed with 200 μg/mL IVIG and detected using anti-IgG antibodies, or with pooled normal mouse sera or pooled normal rabbit sera, each at a 1:1000 dilution, applied onto the array surface and incubated for 1 h in a humidified chamber at 25 °C.
3. After 1 h, remove the slide and immerse it in the TSM Binding Buffer.
4. Incubate the slide with species-specific secondary antibodies (Alexa Fluor 488-anti-IgM or Alexa Fluor 647-anti-IgG) targeting primary IgM or IgG, for 1 h at room temperature.
5. After 1 h, remove the slide and immerse it in the TSM Binding Buffer.
6. Detect fluorescence with a microarray scanner.
7. Determine the integrated spot intensities using Imagene software.

4 Notes

1. Normal human sera were obtained from healthy volunteers through venipuncture. Whole blood was collected into sterile serum tubes according to a protocol approved by the Institutional Review Board (IRB).
2. Mouse and rabbit sera were isolated using venipuncture blood-drawing techniques.
3. A IVIG preparation has been examined from Sandoglobulin (CSL Behring, Bern, Switzerland), which underwent extensive dialysis to remove stabilizing components.
4. Isolated sera were then frozen at −80 °C until further use.
5. Prepare the paper towels by saturating them in water and keep them on the site.
6. Please perform the phenol experiment in a fume hood to ensure proper ventilation and to minimize exposure to harmful fumes. Phenol is toxic and can be hazardous to your health if inhaled, so conducting this procedure in a fume hood is essential for maintaining a safe laboratory environment.
7. The purity of LPS was assessed by SDS-PAGE followed by silver staining. Purified LPS samples of 20 μg, 10 μg, 5 μg, and 2.5 μg were subjected to SDS-PAGE. The resulting gel was washed three times with water, shaken in water for 15 min, and then placed in the destaining buffer (ddH_2O:acetic acid:methanol, 5:1:4) overnight at room temperature. The gel was fixed with 0.7% periodic acid in a destaining buffer (0.7 g sodium periodate in 100 mL of destaining buffer) for 30 min, then washed three times with ddH_2O for 15 min each. Staining reagent (115 mL ddH_2O, 2 mL concentrated ammonium hydroxide, 28 mL 0.1 N sodium hydroxide, 5 mL 20% silver nitrate) was prepared and the gel was stained for 10 min, followed by three 10-minute washes with ddH_2O. Developer solution (100 mL ddH_2O, 5 mg citric acid, 50 μL 37% formaldehyde) was applied until bands appeared. The gel was then washed with ddH_2O, and a photograph was taken to record the results.

Conflicts of Interest

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References

1. Metcalf RA et al (2025) Platelet transfusion: 2025 AABB and ICTMG international clinical practice guidelines. JAMA 334(7):606–617
2. Carson JL et al (2021) Transfusion thresholds for guiding red blood cell transfusion. Cochrane Database Syst Rev 12(12): CD002042
3. Li HY, Guo K (2022) Blood group testing. Front Med (Lausanne) 9:827619
4. Fasano RM, Chou ST (2016) Red blood cell antigen genotyping for sickle cell disease, thalassemia, and other transfusion complications. Transfus Med Rev 30(4):197–201
5. Arthur CM, Stowell SR (2023) The development and consequences of red blood cell Alloimmunization. Annu Rev Pathol 18:537–564
6. Reid ME, Lomas-Francis C, Olsson ML (2012) The blood group antigen factsbook, 3rd edn. Elsevier/AP, Amsterdam. xii, 745 pages
7. Storry JR, Olsson ML (2009) The ABO blood group system revisited: a review and update. Immunohematology 25(2):48–59
8. Jajosky RP et al (2023) ABO blood group antigens and differential glycan expression: perspective on the evolution of common human enzyme deficiencies. iScience 26(1):105798
9. Zimring JC, Spitalnik SL (2015) Pathobiology of transfusion reactions. Annu Rev Pathol 10: 83–110
10. Stowell SR et al (2012) Initiation and regulation of complement during hemolytic transfusion reactions. Clin Dev Immunol 2012: 307093
11. Panch SR, Montemayor-Garcia C, Klein HG (2019) Hemolytic transfusion reactions. N Engl J Med 381(2):150–162
12. Chonat S et al (2020) Eculizumab for complement mediated thrombotic microangiopathy in sickle cell disease. Haematologica 105:2887
13. Chonat S et al (2025) Complement is activated in patients with acute chest syndrome caused by sickle cell disease and represents a therapeutic target. Sci Transl Med 17(807):eadl4922
14. Vichinsky EP et al (1990) Alloimmunization in sickle cell anemia and transfusion of racially unmatched blood. N Engl J Med 322(23): 1617–1621
15. Aygun B et al (2002) Clinical significance of RBC alloantibodies and autoantibodies in sickle cell patients who received transfusions. Transfusion 42(1):37–43
16. Klapper E et al (2010) Toward extended phenotype matching: a new operational paradigm for the transfusion service. Transfusion 50(3): 536–546
17. Pirenne F, Floch A, Diop S (2023) Alloimmunisation against red blood cells in sickle cell disease: transfusion challenges in high-income and low-income countries. Lancet Haematol 10(6):e468–e476
18. Chang DY et al (2023) The ongoing challenge of RBC alloimmunization in the management of patients with sickle cell disease. Presse Med 52(4):104211
19. Zimring JC, Hudson KE (2016) Cellular immune responses in red blood cell alloimmunization. Hematology Am Soc Hematol Educ Program 2016(1):452–456
20. Myle AK, Al-Khattabi GH (2021) Hemolytic disease of the newborn: a review of current trends and prospects. Pediatr Health Med Ther 12:491–498
21. Stowell SR et al (2013) Alloantibodies to a paternally derived RBC KEL antigen lead to hemolytic disease of the fetus/newborn in a murine model. Blood 122(8):1494–1504
22. Verweij E et al (2025) Reporting of management and outcomes in the hemolytic disease of the fetus and newborn: a systematic literature review. J Perinatol 45:1513
23. Kilby MD, Bussel JB, Moise KJ Jr (2025) The contemporary management of haemolytic disease of the fetus and newborn. Vox Sang 120(7):644–652
24. Sugrue RP et al (2024) Maternal red blood cell alloimmunization prevalence in the United States. Blood Adv 8(16):4311–4319
25. Webb J, Delaney M (2018) Red blood cell alloimmunization in the pregnant patient. Transfus Med Rev 32(4):213–219
26. Karafin MS et al (2018) Risk factors for red blood cell alloimmunization in the Recipient Epidemiology and Donor Evaluation Study (REDS-III) database. Br J Haematol 181(5): 672–681
27. Fasano RM et al (2019) Impact of red blood cell antigen matching on alloimmunization and transfusion complications in patients with sickle cell disease: a systematic review. Transfus Med Rev 33(1):12–23
28. Hod EA et al (2008) Cytokine storm in a mouse model of IgG-mediated hemolytic transfusion reactions. Blood 112(3):891–894

29. Liepkalns JS et al (2013) Resistance of a subset of red blood cells to clearance by antibodies in a mouse model of incompatible transfusion. Transfusion 53(6):1319–1327
30. Liepkalns JS et al (2012) Biphasic clearance of incompatible red blood cells through a novel mechanism requiring neither complement nor Fcgamma receptors in a murine model. Transfusion 52(12):2631–2645
31. Arthur CM et al (2019) Examining the role of complement in predicting, preventing, and treating hemolytic transfusion reactions. Transfus Med Rev 33(4):217–224
32. Arthur CM et al (2021) Antigen density dictates RBC clearance, but not antigen modulation, following incompatible RBC transfusion in mice. Blood Adv 5(2):527–538
33. Branch DR (2015) Anti-A and anti-B: what are they and where do they come from? Transfusion 55(Suppl 2):S74–S79
34. Yazdanbakhsh K (2016) Immunoregulatory networks in sickle cell alloimmunization. Hematology Am Soc Hematol Educ Program 2016(1):457–461
35. Yazdanbakhsh K, Ware RE, Noizat-Pirenne F (2012) Red blood cell alloimmunization in sickle cell disease: pathophysiology, risk factors, and transfusion management. Blood 120(3): 528–537
36. Bao W et al (2009) Regulatory T-cell status in red cell alloimmunized responder and nonresponder mice. Blood 113(22):5624–5627
37. Jajosky RP et al (2024) Dynamics of antibody engagement of red blood cells in vivo and in vitro. Front Immunol 15:1475470
38. Jajosky RP et al (2025) CD47 regulates antigen modulation and red blood cell clearance following an incompatible transfusion. Front Immunol 16:1548548
39. Mener A et al (2019) Antibody-mediated immunosuppression can result from RBC antigen loss independent of Fcgamma receptors in mice. Transfusion 59(1):371–384
40. Mener A et al (2018) Complement component 3 negatively regulates antibody response by modulation of red blood cell antigen. Front Immunol 9:676
41. Thomas TA et al (2023) Reticulocytes in donor blood units enhance red blood cell alloimmunization. Haematologica 108(10):2639–2651
42. Jash A et al (2024) Complement C3 and marginal zone B cells promote IgG-mediated enhancement of RBC alloimmunization in mice. J Clin Invest 134(8):e167665
43. Jajosky R et al (2023) Prior immunization against an intracellular antigen enhances subsequent red blood cell alloimmunization in mice. Blood 141(21):2642–2653
44. Zerra PE et al (2021) Marginal zone B cells mediate a CD4 T-cell-dependent extrafollicular antibody response following RBC transfusion in mice. Blood 138(8):706–721
45. Mener A et al (2018) Complement serves as a switch between CD4+ T cell-independent and -dependent RBC antibody responses. JCI Insight 3(22):e121631
46. Patel SR et al (2018) Recipient priming to one RBC alloantigen directly enhances subsequent alloimmunization in mice. Blood Adv 2(2): 105–115
47. Patel SR et al (2018) Marginal zone B cells induce alloantibody formation following RBC transfusion. Front Immunol 9:2516
48. Gibb DR et al (2017) Type I IFN is necessary and sufficient for inflammation-induced red blood cell alloimmunization in mice. J Immunol 199(3):1041–1050
49. Gibb DR et al (2017) B cells require Type 1 interferon to produce alloantibodies to transfused KEL-expressing red blood cells in mice. Transfusion 57(11):2595–2608
50. Arthur CM et al (2017) Antigen density dictates immune responsiveness following red blood cell transfusion. J Immunol 198(7): 2671–2680
51. Couvidou A et al (2023) Marginal zone B cells are responsible for the production of alloantibodies following platelet transfusion in mice. Blood Adv 7(8):1356–1365
52. Zerra PE et al (2017) Marginal zone B cells are critical to factor VIII inhibitor formation in mice with hemophilia a. Blood 130(23): 2559–2568
53. Arthur CM et al (2022) Nonhuman glycans can regulate anti-factor VIII antibody formation in mice. Blood 139(9):1312–1317
54. Arthur CM et al (2016) CD8+ T cells mediate antibody-independent platelet clearance in mice. Blood 127(14):1823–1827
55. Arthur CM et al (2022) Clodronate inhibits alloimmunization against distinct red blood cell alloantigens in mice. Transfusion 62(5): 948–953
56. Zimring JC et al (2005) Nonhemolytic antibody-induced loss of erythrocyte surface antigen. Blood 106(3):1105–1112
57. Sullivan HC et al (2017) Daratumumab (anti-CD38) induces loss of CD38 on red blood cells. Blood 129(22):3033–3037
58. Girard-Pierce KR et al (2013) A novel role for C3 in antibody-induced red blood cell clearance and antigen modulation. Blood 122(10): 1793–1801
59. Jajosky RP et al (2023) Antibody-mediated antigen loss switches augmented immunity to

antibody-mediated immunosuppression. Blood 142(12):1082–1098

60. Adam I et al (2025) ABO-A antibody induction in mice is T cell-dependent, estrogen-independent, and modulated by CD22. Am J Transplant 25(6):1180–1192
61. Adam I et al (2023) Sex, T cells, and the microbiome in natural ABO antibody production in mice. Transplantation 107(11):2353–2363
62. Stowell CP, Stowell SR (2019) Biologic roles of the ABH and Lewis histo-blood group antigens part I: infection and immunity. Vox Sang 114(5):426–442
63. Hale RC et al (2024) The role of glycosylation in clinical allergy and immunology. J Allergy Clin Immunol 153(1):55–66
64. Jan HM et al (2024) Galectin-4 antimicrobial activity primarily occurs through its C-terminal domain. Mol Cell Proteomics 23(5):100747
65. Arthur CM, Cummings RD, Stowell SR (2014) Using glycan microarrays to understand immunity. Curr Opin Chem Biol 18: 55–61
66. Blenda AV et al (2022) Galectin-9 recognizes and exhibits antimicrobial activity toward microbes expressing blood group-like antigens. J Biol Chem 298(4):101704
67. Wu SC et al (2022) Innate immune Galectin-7 specifically targets microbes that decorate themselves in blood group-like antigens. iScience 25(7):104482
68. Wu SC et al (2021) The SARS-CoV-2 - receptor-binding domain preferentially recognizes blood group a. Blood Adv 5(5): 1305–1309
69. Wu SC et al (2021) Full-length Galectin-3 is required for high affinity microbial interactions and antimicrobial activity. Front Microbiol 12: 731026
70. Stowell SR et al (2014) Microbial glycan microarrays define key features of host-microbial interactions. Nat Chem Biol 10(6): 470–476
71. Wu SC et al (2023) Blood group A enhances SARS-CoV-2 infection. Blood 142(8): 742–747
72. Liu FT, Stowell SR (2023) The role of galectins in immunity and infection. Nat Rev Immunol 23(8):479–494
73. Luetscher RND et al (2020) Unique repertoire of anti-carbohydrate antibodies in individual human serum. Sci Rep 10(1):15436
74. Rillahan CD, Paulson JC (2011) Glycan microarrays for decoding the glycome. Annu Rev Biochem 80:797–823
75. Stowell SR et al (2008) Galectin-1, -2, and -3 exhibit differential recognition of sialylated glycans and blood group antigens. J Biol Chem 283(15):10109–10123
76. Stowell SR et al (2008) Dimeric Galectin-8 induces phosphatidylserine exposure in leukocytes through polylactosamine recognition by the C-terminal domain. J Biol Chem 283(29): 20547–20559
77. Ho AD et al (2022) An automated approach to assess relative galectin-glycan affinity following glycan microarray analysis. Front Mol Biosci 9: 893185
78. Stowell SR et al (2009) Ligand reduces galectin-1 sensitivity to oxidative inactivation by enhancing dimer formation. J Biol Chem 284(8):4989–4999
79. Tulin EKC et al (2025) Blood group O expression in normal tissues and tumors. FEBS J 292: 5697
80. Jia T et al (2025) Novel click coupling chemistry to explore glycan recognition. ACS Cent Sci 11(5):753–769
81. Stowell SR et al (2008) Differential roles of galectin-1 and galectin-3 in regulating leukocyte viability and cytokine secretion. J Immunol 180(5):3091–3102
82. Stowell SR et al (2010) Innate immune lectins kill bacteria expressing blood group antigen. Nat Med 16(3):295–301